Lecture Scribe CSE400

L18.pdf
PDF

Act as an expert scribe writer specifically for Lectures taken by professors of a B.Tech(CSE department) , so i want you to generate a well structured lecture scribe which should include definition and notation , assumption and conditions , statements of theorem or results , proofs or proof sketches(step by step),worked examples with intermediate steps and clear logical flow between ideas and be very sure that it should not be like a creative explaination,simplified tutorial,textbook rewrite,solution manual for unseen problems and a place to add intution not discusses in class and the main purpose for the scribe is faithful,exam ready reconstruction of what was taught in class, written so that a student can later rely on it as reference material while studying and it should allow to recall definations and notations, reconstruct proofs and derivations, follow example solution step by step and understand logical dependencies between concepts and it should basically answer a question that if i study only this document , can i reliably revise what was taught and as it will be used for exam reference you must prioritize correctness, include explicit reasoning when appropriate and follow the lecture's structure and scope (which i have uploaded in pdf for you context) and you must not invent content by yourself and for the reference format to follow i am giving you in text follow that absolutely "Lecture Scribe Example Structure
Course: Fundamentals of Probability in Computing
Lecture Topic: Conditional Probability and Independence
Date: [Insert Date]
Scribe: [Your Name]

1. Opening Context (Real-Life Connection)
(If mentioned in lecture pdf)

2. Definitions and Notation
2.1 Conditional Probability
Definition (as stated in lecture):
For events A and B where P(B) > 0, the conditional probability of A given B is:
P(A|B)=P(A∩B)P(B)P(A|B) = \frac{P(A \cap B)}{P(B)}P(A|B)=P(B)P(A∩B)
Notation introduced:

P(A|B): "probability of A given B"
A ∩ B: "A and B both occur"
The condition P(B) > 0 is necessary (cannot condition on impossible events)

3. Key Theorem: Law of Total Probability
3.1 Statement

Theorem (as stated in lecture):
Let $B_1, B_2, ..., B_n$ be a partition of the sample space Ω (mutually exclusive and exhaustive). Then for any event A:
P(A)=∑i=1nP(A|Bi)·P(Bi)P(A) = \sum_{i=1}^{n} P(A|B_i) \cdot P(B_i)P(A)=i=1∑nP(A|Bi)·P(Bi)
3.2 Conditions/Assumptions

$\{B_1, B_2, ..., B_n\}$ must partition Ω
$P(B_i) > 0$ for all i
Mutually exclusive: $B_i \cap B_j = \emptyset$ for $i \neq j$
Exhaustive: $B_1 \cup B_2 \cup ... \cup B_n = \Omega$

3.3 Proof (as presented in class)
Step 1: Since $\{B_1, ..., B_n\}$ partition Ω, we can write:

A=A∩Ω=A∩(B1∪B2∪...∪Bn)A = A \cap \Omega = A \cap (B_1 \cup B_2 \cup ... \cup B_n)A=A∩Ω=A∩(B1∪B2∪...∪Bn)
Step 2: By distributivity:

A=(A∩B1)∪(A∩B2)∪...∪(A∩Bn)A = (A \cap B_1) \cup (A \cap B_2) \cup ... \cup (A \cap B_n)A=(A∩B1)∪(A∩B2)∪...∪(A∩Bn)
Step 3: Since B_i are mutually exclusive, $(A \cap B_i)$ are also mutually exclusive.
Step 4: By additivity of probability:

P(A)=P(A∩B1)+P(A∩B2)+...+P(A∩Bn)P(A) = P(A \cap B_1) + P(A \cap B_2) + ... + P(A \cap B_n)P(A)=P(A∩B1)+P(A∩B2)+...+P(A∩Bn)
Step 5: Apply definition of conditional probability $P(A \cap B_i) = P(A|B_i) \cdot P(B_i)$:

P(A)=∑i=1nP(A|Bi)·P(Bi)P(A) = \sum_{i=1}^{n} P(A|B_i) \cdot P(B_i)P(A)=i=1∑nP(A|Bi)·P(Bi)
4. What This Theorem Actually Does
In plain terms (from lecture):
When we don't know P(A) directly but know:

How to partition the sample space into scenarios $B_1, B_2, ..., B_n$
The probability of each scenario $P(B_i)$
How likely A is within each scenario $P(A|B_i)$

Then we can compute P(A) by weighted average across all scenarios.
Where it helps:

Computing probabilities in multi-stage processes
Network reliability analysis (multiple failure modes)
Error analysis in algorithms (different input cases)

5. Worked Example (from lecture)
Problem (Example 3.2 from slides):
A software testing system has three bug detection tools:

Tool 1: detects 40% of bugs, used 50% of the time
Tool 2: detects 60% of bugs, used 30% of the time
Tool 3: detects 80% of bugs, used 20% of the time

A bug is detected. What's the probability it was found by Tool 3?
Solution (step-by-step as shown):
Step 1: Define events

T_1, T_2, T_3: tool 1, 2, or 3 is used
D: bug is detected

Step 2: List given probabilities

$P(T_1) = 0.5$, $P(T_2) = 0.3$, $P(T_3) = 0.2$
$P(D|T_1) = 0.4$, $P(D|T_2) = 0.6$, $P(D|T_3) = 0.8$

Step 3: Find P(D) using Law of Total Probability

P(D)=P(DIT1)·P(T1)+P(DIT2)·P(T2)+P(DIT3)·P(T3)P(D) = P(D|T_1) \cdot P(T_1) + P(D|T_2) \cdot P(T_2) + P(D|T_3) \cdot P(T_3)P(D)=P(DIT1)·P(T1)+P(DIT2)·P(T2)+P(DIT3)·P(T3)
P(D)=(0.4)(0.5)+(0.6)(0.3)+(0.8)(0.2)P(D) = (0.4)(0.5) + (0.6)(0.3) + (0.8)(0.2)P(D)=(0.4)(0.5)+(0.6)(0.3)+(0.8)(0.2)
P(D)=0.2+0.18+0.16=0.54P(D) = 0.2 + 0.18 + 0.16 = 0.54P(D)=0.2+0.18+0.16=0.54
Step 4: Apply Bayes' Theorem (we want $P(T_3|D)$)

P(T3ID)=P(DIT3)·P(T3)P(D)P(T_3|D) = \frac{P(D|T_3) \cdot P(T_3)}{P(D)}P(T3ID)=P(D)P(DIT3)·P(T3)
P(T3ID)=(0.8)(0.2)0.54=0.160.54≈0.296P(T_3|D) = \frac{(0.8)(0.2)}{0.54} = \frac{0.16}{0.54} \approx 0.296P(T3ID)=0.54(0.8)(0.2)=0.540.16≈0.296
Answer: Approximately 29.6% chance the detection came from Tool 3.

6. Definition: Independence
6.1 Statement
Definition (as stated in lecture):
Events A and B are *independent* if and only if:

P(A∩B)=P(A)·P(B)P(A \cap B) = P(A) \cdot P(B)P(A∩B)=P(A)·P(B)
Equivalent condition (proven in class):
If $P(B) > 0$, then A and B are independent $\iff P(A|B) = P(A)$

7. Connection to Previous Material
Link to Lecture 2 (Basic Probability Axioms):

Today's $P(A|B)$ definition uses axiom $P(\Omega) = 1$ restricted to B as new sample space
Law of Total Probability generalizes the addition rule for mutually exclusive events

Foundation for next lecture:

8. Summary & Key Takeaways (from lecture conclusion)
Main points emphasized:

Conditional probability updates our beliefs given new information
Law of Total Probability breaks hard problems into cases
Independence is about informational irrelevance, not correlation
Bayes' Theorem (mentioned briefly) inverts conditional probabilities

APPENDIX: Quick Reference Sheet
Formulas
ConceptFormulaConditionConditional Probability$P(A|B) = P(A \cap B) / P(B)$$P(B) > 0$Law of Total Probability$P(A) = \Sigma_i P(A|B_i)\cdot P(B_i)$$\{B_i\}$ partition ΩIndependenceP(A ∩ B) = P(A)·P(B)AlwaysBayes' Theorem$P(A|B) = P(B|A)\cdot P(A) / P(B)$$P(B) > 0$
Notation

$P(A|B)$: Probability of A given B
$A \cap B$: Both A and B occur
$A \cup B$: At least one of A or B occurs
Ω: Sample space (all possible outcomes)
$\varnothing$: Empty set (impossible event)
$\{B_1, ..., B_n\}$ partition Ω: Mutually exclusive and exhaustive

Key Conditions to Remember
✓ For $P(A|B)$ to exist: $P(B) > 0$
✓ For Law of Total Probability: $\{B_i\}$ must partition Ω
✓ Partition means: mutually exclusive ($B_i \cap B_j = \varnothing$) and exhaustive ($\cup B_i = \Omega$)
Buzz Words from Lecture

"Given" → conditioning, restriction of sample space
"Partition" → mutually exclusive + exhaustive
"Independent" → no informational relevance
"Weighted average" → how professor described Law of Total Probability

Common Tricks

Finding $P(A|B)$ when given $P(B|A)$: Use Bayes' Theorem
Complex P(A) calculation: Try Law of Total Probability with natural partition
Testing independence: Check if $P(A|B) = P(A)$ OR $P(A \cap B) = P(A)\cdot P(B)$

Important Warnings (from lecture)
⚠️ $P(A|B) \neq P(B|A)$ in general
⚠️ "Mutually exclusive" and "independent" are different concepts
⚠️ Cannot condition on events with probability zero

End of Scribe"the audience are my fellow classmates currently in 2nd year of B.Tech(CSE) so give accordigly and strictly follow all given instruction

Here is your **faithful, exam-ready lecture scribe** strictly reconstructed from the uploaded lecture slides , following your exact structure and constraints.

Lecture Scribe

Course: CSE400 – Fundamentals of Probability in Computing
Lecture Topic: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables
Date: March 12, 2026
Scribe: Het Patel

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1. Opening Context (Real-Life Connection)

(Explicitly mentioned in lecture slides)

- Joint Gaussian models are widely used in:
  - Signal processing
  - Communication systems
  - Modeling noisy real-world measurements

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2. Marginal PMF

2.1 Definition

Definition (as stated in lecture):

For discrete random variables X and Y, the joint PMF is:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The marginal PMFs are:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea (from lecture):
The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable** .

2.2 Notation

- $p_{XY}(x, y)$: Joint PMF
- $p_X(x), p_Y(y)$: Marginal PMFs
- Summation over other variable removes dependency

2.3 Marginal PMF from Joint PMF Table

(From table shown in lecture slides)

- Row sums → $p_X(x)$
- Column sums → $p_Y(y)$

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

2.4 Worked Example (from lecture slides)

Given Joint PMF:

	$y=0$	$y=1$
$x=0$	$\frac{1}{8}$	$\frac{1}{4}$
$x=1$	$\frac{3}{8}$	$\frac{1}{4}$

Step 1: Compute $p_X(x)$

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

Step 2: Compute $p_Y(y)$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Answer:

$$p_X(x) = \begin{cases} \frac{3}{8}, & x=0 \\ \frac{5}{8}, & x=1 \end{cases} \quad p_Y(y) = \begin{cases} \frac{1}{2}, & y=0 \\ \frac{1}{2}, & y=1 \end{cases}$$

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3. Correlation (Raw Measure)

3.1 Definition (from lecture)

$$R_{XY} = E[XY]$$

3.2 Interpretation

- Measures **average value of product** XY
- Computed **about origin (not centered)**
- Depends on:

- Mean
- Spread
- Dependence

3.3 Important Observation

- $R_{XY} = 0 \Rightarrow$ orthogonal about origin
- Not a clean dependence measure

Transition (as emphasized in lecture):
To remove mean effects → leads to **Covariance**

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4. Covariance

4.1 Definition

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where:

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

4.2 Algebraic Form (Derived in Lecture)

Step-by-step expansion:

$$\begin{aligned}\mathrm{Cov}(X, Y) &= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y] \\ &= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y \\ &= E[XY] - \mu_X\mu_Y\end{aligned}$$

4.3 Final Formula

$$\boxed{\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

4.4 Interpretation (from slides)

- Positive covariance → variables increase together
- Negative covariance → one increases, other decreases
- Zero covariance → no linear relationship

⚠️ Magnitude depends on units → cannot measure strength

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5. Properties of Covariance

1. **Symmetry**

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

2. **Variance as Special Case**

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X)$$

3. **Linearity**

$$\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$$

4. **Independence ⇒ Zero Covariance**

$$X \perp Y \Rightarrow \mathrm{Cov}(X,Y) = 0$$

⚠️ **Important Warning (from lecture):**
Zero covariance ≠ independence

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6. Worked Example 1 (from lecture)

Problem:
If $Y = -2X + 3$, find $\mathrm{Cov}(X,Y)$

Solution

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

Step 1: Compute XY

$$XY = X(-2X+3) = -2X^2 + 3X$$

Step 2: Compute expectations

$$E[XY] = E[-2X^2 + 3X] = -2E[X^2] + 3E[X]$$

$$E[Y] = E[-2X+3] = -2E[X] + 3$$

Step 3: Substitute

$$\begin{aligned}\mathrm{Cov}(X,Y) &= (-2E[X^2] + 3E[X]) - E[X](-2E[X]+3)\\ &= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X]\\ &= -2E[X^2] + 2(E[X])^2\end{aligned}$$

Step 4: Use variance identity

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

$$\Rightarrow \mathrm{Cov}(X,Y) = -2\mathrm{Var}(X)$$

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7. Pearson's Correlation Coefficient

7.1 Definition

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

7.2 Properties

- $-1 \leq \rho_{XY} \leq 1$
- Unit-free
- Measures **strength + direction**

7.3 Interpretation

- $\rho > 0$: Positive linear trend
- $\rho < 0$: Negative linear trend
- $\rho = 0$: No linear correlation

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8. Worked Example 2 (Continuous Case)

Given:

$$f_{XY}(x,y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4,\ 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

Key Results (as derived in lecture slides)

$$E[X] = \frac{8}{3}, \quad E[Y] = \frac{31}{9}$$

$$E[XY] = \frac{248}{27}$$

$$E[2X + 3Y] = \frac{47}{3}$$

$$\text{Var}(X) = \frac{8}{9}, \quad \text{Var}(Y) = \frac{92}{81}$$

Covariance

$$\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

$$= \frac{248}{27} - \left(\frac{8}{3} \cdot \frac{31}{9}\right) = 0$$

Correlation

$$\rho_{XY} = 0$$

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9. Joint Gaussian Random Variables

9.1 Definition

A pair (X, Y) is jointly Gaussian if:

- Joint PDF has **bivariate Gaussian form**
- Marginals are Gaussian

9.2 Joint PDF (as given in lecture)

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\right.\right.$$

9.3 Parameters

- μ_X, μ_Y: Means
- σ_X, σ_Y: Standard deviations
- ρ: Correlation coefficient

9.4 Interpretation (from lecture example)

- If $\rho > 0$: variables increase together
- Example given:
  - CO_2 concentration vs temperature
  - Higher CO_2 → higher temperature trend

10. Connection to Previous Material

- Uses:
  - Expectation $E[X]$
  - Variance definition
- Extends:
  - Single RV → Joint RV analysis
- Leads to:
  - Multivariate distributions

11. Summary & Key Takeaways

- Marginal PMF = sum over other variable
- $E[XY]$ is raw correlation (not centered)
- Covariance removes mean effect
- Covariance:
  - sign → direction
  - not strength
- Pearson coefficient:
  - normalized measure
  - gives strength + direction
- Zero covariance ≠ independence
- Joint Gaussian:
  - fundamental continuous model

APPENDIX: Quick Reference Sheet

Formulas

Concept	Formula
Marginal PMF	$p_X(x) = \sum_y p_{XY}(x, y)$
Correlation (raw)	$E[XY]$
Covariance	$E[XY] - E[X]E[Y]$
Pearson Coefficient	$\rho = \frac{\text{Cov}}{\sigma_X \sigma_Y}$
Joint Gaussian PDF	(as given above)

Key Conditions

✓ Marginalization → sum over other variable
✓ Covariance uses centered variables
✓ Independence ⇒ covariance = 0
⚠️ Converse not true

Buzz Words (from lecture)

- "Summing out" → marginalization
- "Centered variables" → covariance
- "Normalized" → Pearson coefficient
- "Linear relationship" → correlation

Space for Intuition (Not in Lecture)

- Marginal PMF = "ignore the other variable"
- Covariance = "do deviations move together?"
- Pearson = "scaled covariance"

✅ This scribe is **fully aligned with lecture slides**, structured for **exam recall, derivation reconstruction, and stepwise problem solving**.